

Rethinking NLP Evaluation in the Age of LLMs

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Overview of the Seminar

- **A bit of context in LLM evaluation**
- **LLM-leaderboard vs LLM-Arena**
- **Single prompt vs multiple prompts**
- **Multiple-choice vs open-ended tasks**
- **LLM-as-Judge**
- **More issues...**

Conclusion

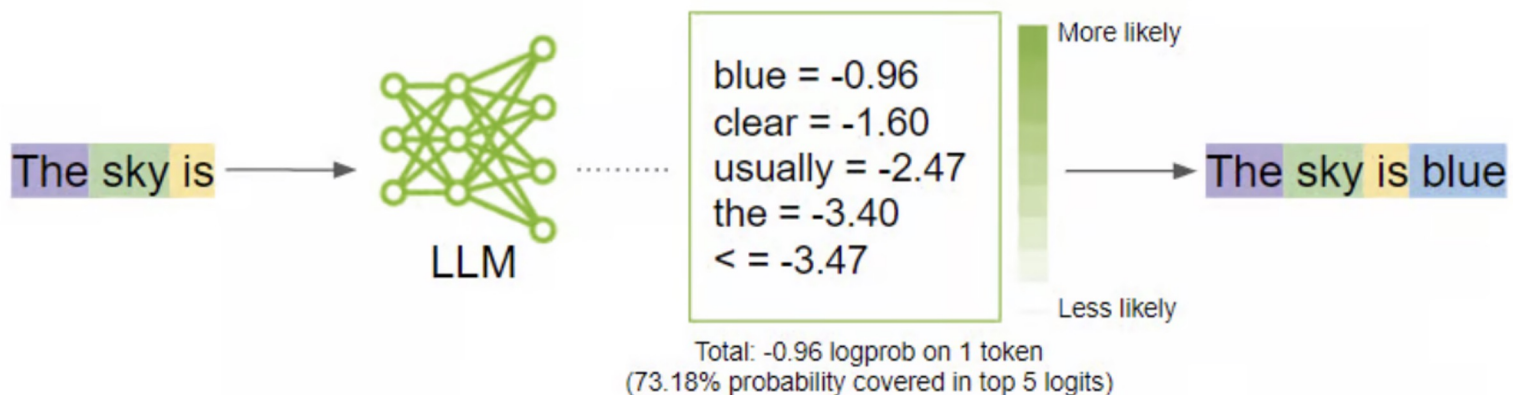
- A fair evaluation of LLMs is still a research challenge

A Bit of Context in NLP Evaluation

NLP with LLMs

Large Language Models (LLMs)

- Systems trained to predict the probability of the next word (**next word prediction**) given the context around that word
- LLMs are **generative models** because they produce new textual content
- Recent LLMs **simulate linguistic skills** of a speaker:
 - conversing, answering questions, giving advice, making summaries, writing a text on a certain topic, translating from one language to another, giving explanations, etc.

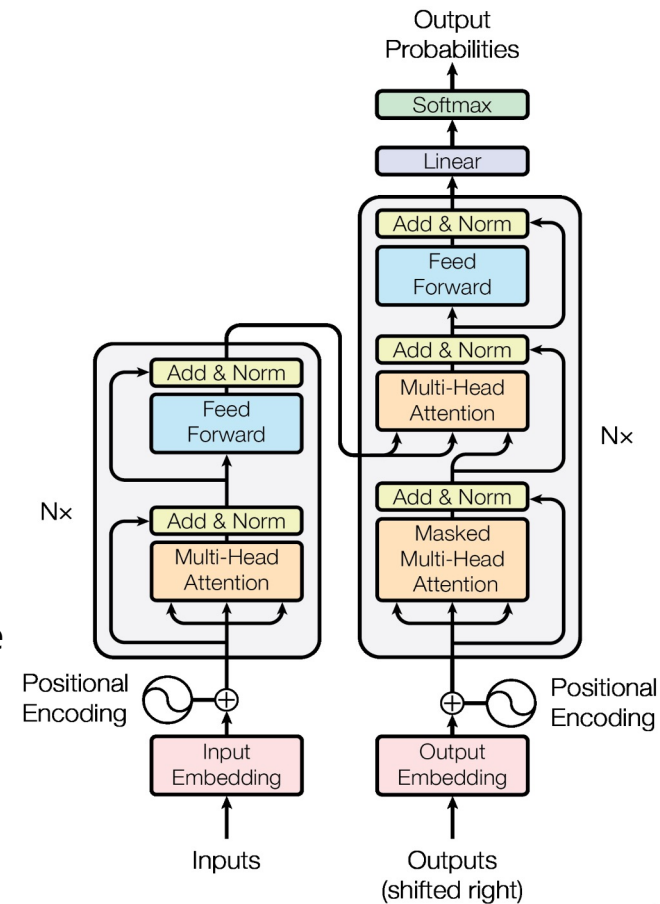


Source: Annie Surla / Nvidia Blog

LLMs: Technological Recipe

Main technologies behind LLMs

- **Continuous representation of words:** the meaning of words is compressed into numerical vectors (**word embeddings**)
- **Transformers and attention:** word embeddings are modified by neural architectures to adapt them to the context of use (LLM input).
- **Training data:** as many text as possible are used during the learning phase, so as to cover most contexts of use, resulting in the LLM parameters (the model).



NLP Evaluation

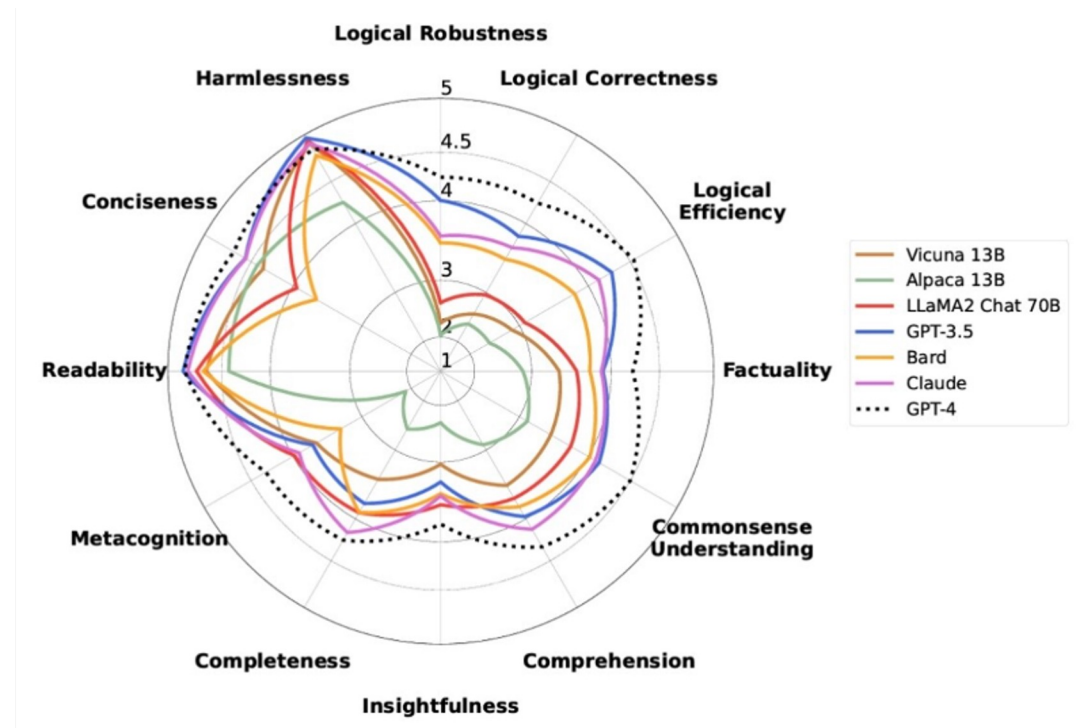
Benchmarking: fundamental for boosting technological progress

Goals

- Compare LLMs with different characteristics
- Improve existing LLMs
- Orient research directions
- Test new applications

Key ingredients

- Task definition (input, output)
- Dataset: dev, train, test (annotations)
- Metrics



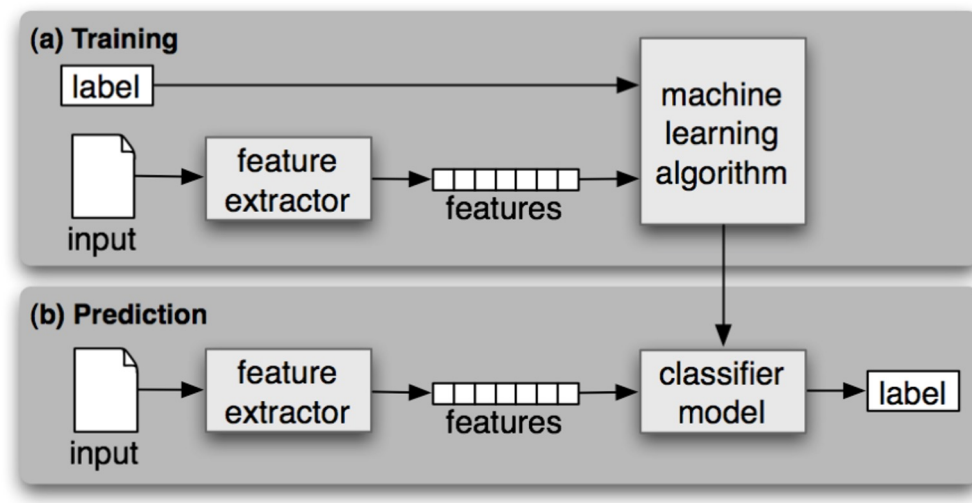
Source: Seonghyeon Ye et al: Flask: Fine-grained Language Model Evaluation based on Alignment Skill Sets, ICLR, 2024.

Evaluation with Traditional ML

SVM: *Support Vector Machines* for NLP (Mullen et al. 2004)

Method

- The model (SVM) does not have any prior knowledge
- **Feature engineering** on training data (a vector of features for each data point)
- One model (supervised) – one task



Source:
<https://www.nltk.org/book/ch06.html>

Evaluation with BERT

BERT: *Bidirectional Encoder Representations from Transformers* (J. Devlin et al. 2018)

Method

- BERT **contextual embeddings are good features** (no need of feature engineering)!
- A classifier is trained for prediction on **annotated data** using embedding as features
- The model does not change, but **a new classifier for each task is needed**

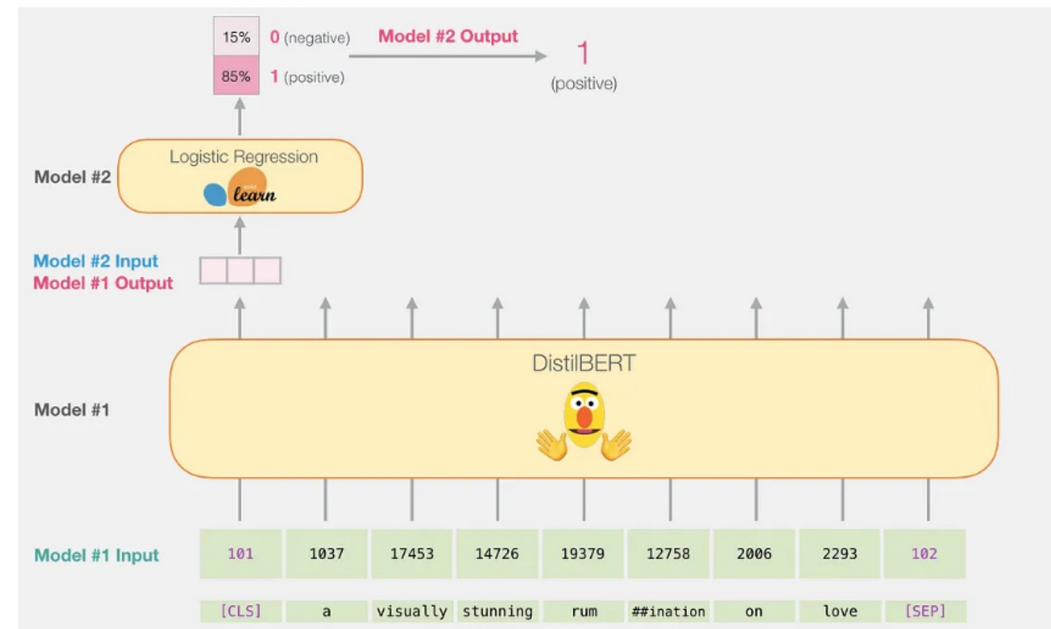


Image by Huggingface

Evaluation with LLMs (Fine-tuned)

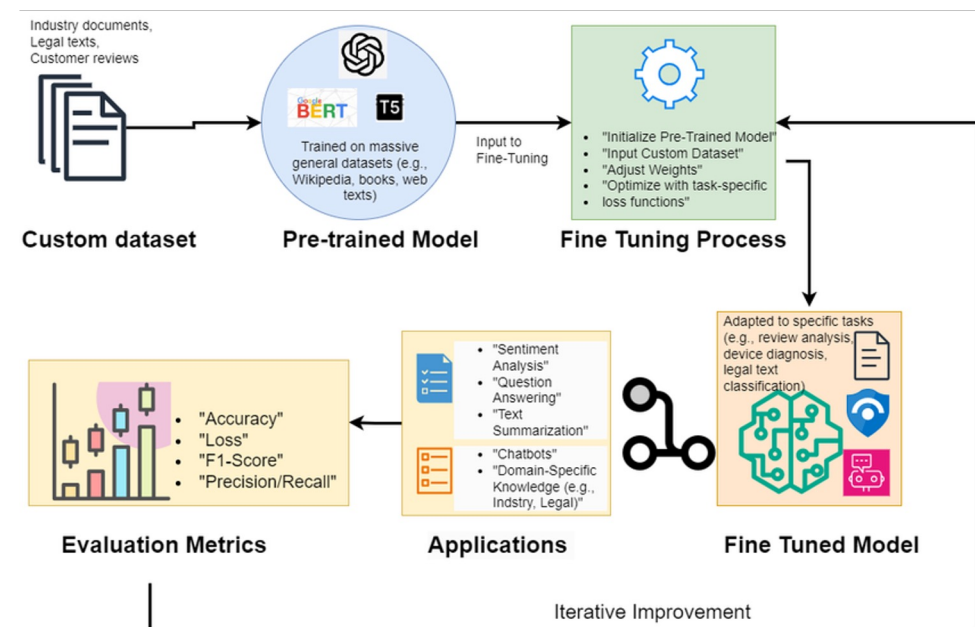
LLMs (e.g., Llama family) starting from 2023 are pre-trained and then instruction-tuned

Method

- Training data used for fine-tuning the LLM
- During fine-tuning on a specific downstream task, LLM parameters are updated
- Typically, forgetting after fine-tuning
- **Prompt engineering** is needed
- Training data usually **do not have prompts**, need to be adapted to LLM interactive style

E.g. *tweet* = POSITIVE

Which is the sentiment of *tweet*? Possible answers: POSITIVE, NEGATIVE



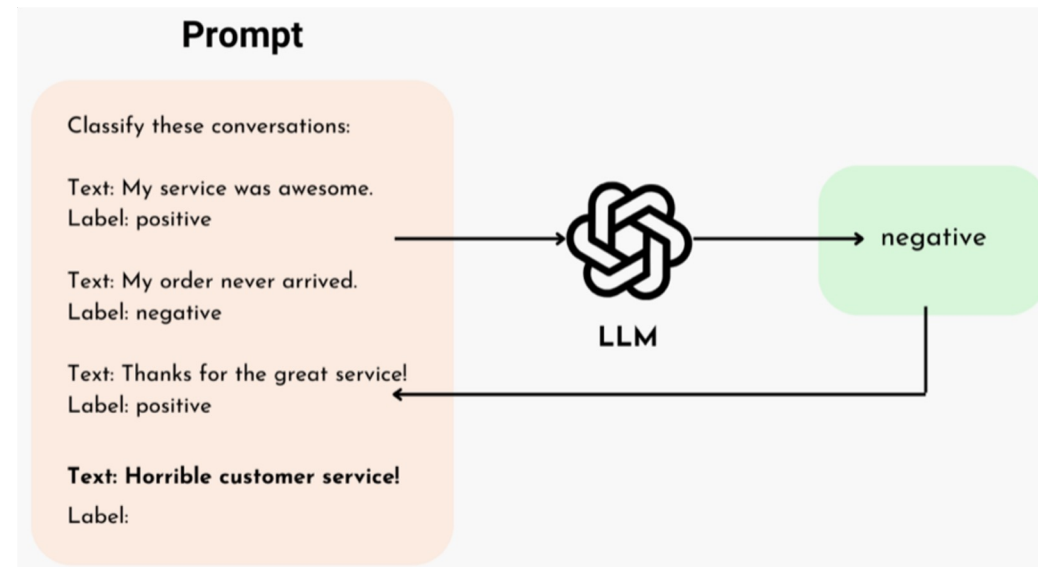
Source: E. Zeydan et al., 2024

Evaluation with LLMs (zero and few shot)

LLMs: large pre-trained and instructed models

Method

- **No need for training data**, or just a few!
- Model **parameters are kept frozen**: a multitude of downstream tasks!
- **Prompt engineering** is needed
- LLMs are generative models: **output needs to be parsed**

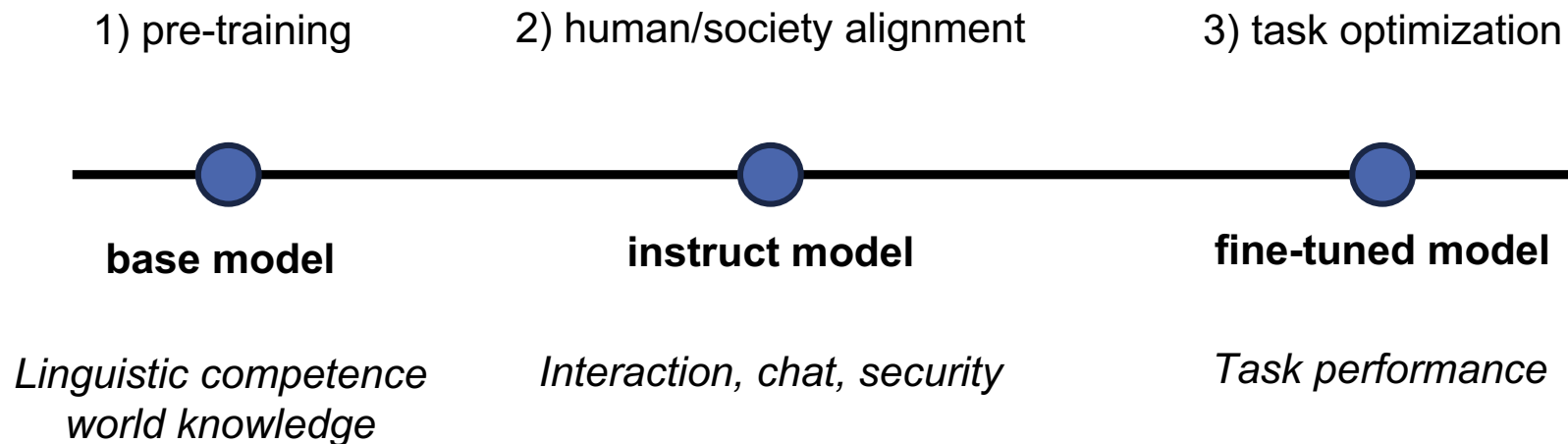


Source: <https://datasciencedojo.com/blog/dynamic-few-shot-prompting/>

LLMs: What to Evaluate (1)

Three sequential phases of LLM development

At each phase new abilities are added and, possibly, previous abilities are weakened

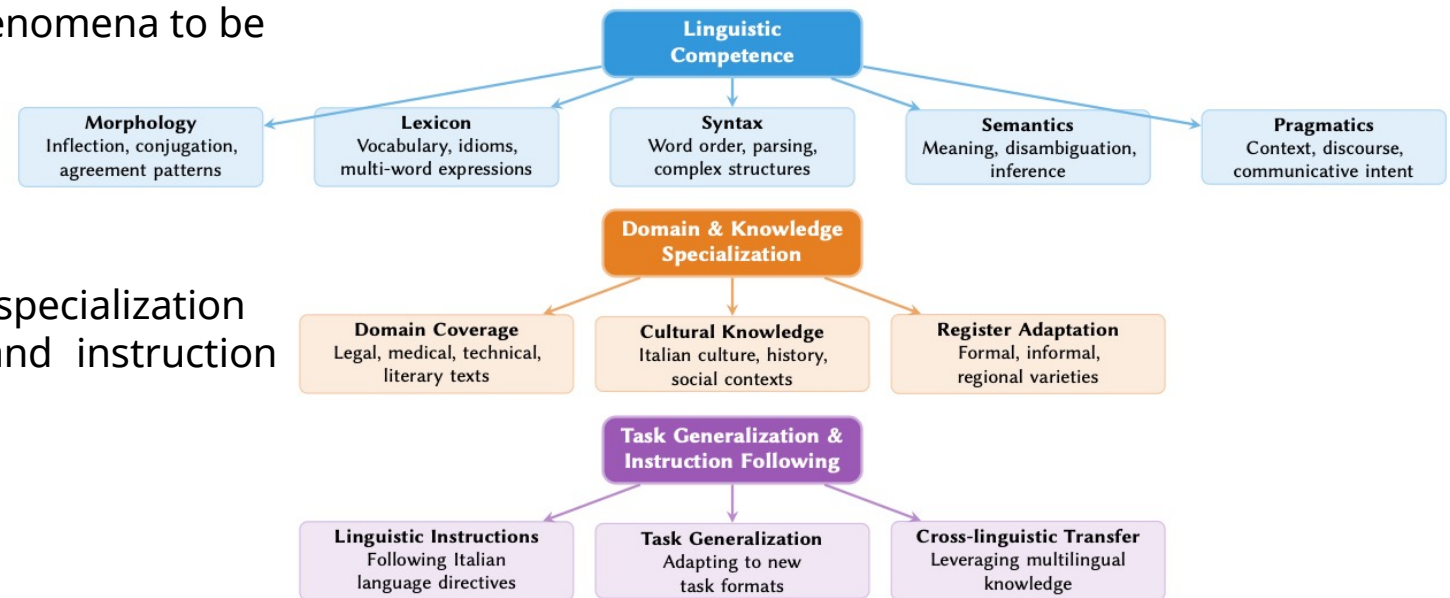


LLMs: What to Evaluate (2)

An ontology of linguistic phenomena to be covered.

Three levels

- Linguistic competence
- Domain & knowledge specialization
- Task generalization and instruction following



[Source: Moroni et al., 2025](#)

LLMs: What to Evaluate (2)

A focus on the current situation for Italian

Three kind of dataset

- Translated from English
- Native Italian (e.g. from the Evalita initiative)
- Collections of tasks
- For some phenomena the coverage on current dataset is still limited

Dataset	Morphology	Lexical	Syntax	Semantics	Pragmatics	Domain	Culture	Register	Ling. Instr.	Task Gen.	Cross-Ling.
AI2-ARC	X	X	X	✓	X	✓	X	X	X	X	X
BoolQ	X	✓	X	✓	X	✓	X	X	X	X	X
GSM8K	X	X	X	X	X	✓	X	X	X	X	X
HellaSwag	X	✓	X	✓	✓	X	X	X	X	X	X
MMLU	X	✓	X	✓	X	✓	X	X	X	X	X
PIQA	X	✓	X	✓	X	X	X	X	X	X	X
SciQ	X	X	X	✓	X	✓	X	X	X	X	X
TruthfulQA	X	X	X	X	X	✓	X	X	X	X	X
WinoGrande	X	✓	X	✓	✓	X	X	X	X	X	X
Admission Test	X	✓	X	✓	X	✓	X	X	X	X	X
AMI 2020	X	✓	X	✓	X	X	✓	X	X	X	X
CLinkaRT 2023	X	✓	X	✓	X	✓	X	X	X	X	X
DiscoTEX	X	X	✓	✓	✓	X	X	X	X	X	X
GhigliottinAI	X	✓	X	✓	X	X	✓	X	X	X	X
HaSpeeDe2	X	✓	X	✓	✓	X	✓	X	X	X	X
LexSub	X	✓	X	✓	X	X	X	X	X	X	X
NERMUD	X	✓	X	✓	X	X	X	X	X	X	X
PreLearn20	X	X	X	✓	X	✓	X	X	X	X	X
PreTENS 22	X	✓	X	✓	X	X	X	X	X	X	X
QA4FAQ	X	X	X	✓	X	✓	X	X	X	X	X
QuandHo	X	X	X	✓	X	✓	✓	X	X	X	X
SENTIPOLC	X	✓	X	✓	✓	X	X	X	X	X	X
Sum-FP	X	X	X	✓	X	✓	X	X	✓	X	X
Textual Entailment	X	X	X	✓	X	X	X	X	X	X	X
WiC-ITA	X	✓	X	✓	X	X	X	X	X	X	X
ITA-Bench	X	✓	X	✓	✓	✓	✓	X	X	X	✓
EvalITA-LLM	X	✓	X	✓	✓	✓	✓	X	✓	X	X
ITALIC	✓	✓	✓	✓	X	✓	✓	X	X	X	X

Source: Moroni et al., 2025

Evaluating LLMs: Metrics

Evaluation Metrics

Classification Tasks (multiple choice)

- Metrics: accuracy, precision, recall, F1, etc.

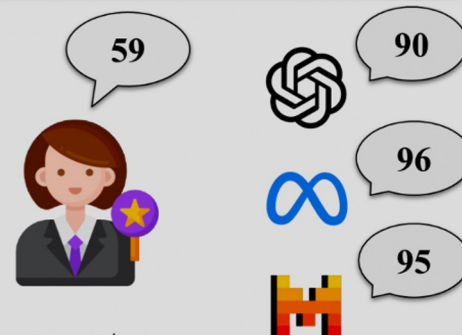
Generation Tasks

- Many benchmarks include "references" (e.g. translations, summaries) developed by experts
- Metrics based on similarity between reference and model output (e.g. bleu, rouge, bert-score, etc.): limited effectiveness
- **LLM-as-a-judge**: an LLM compares reference and model output. Approach to be consolidated.

WMT 2023 - EnDe

Instruction: Your task is to evaluate the quality of machine translation output on a scale from 0 to 100 [...]. Evaluation Criteria: [...]

Source: Great backpack but overkill on the straps
Reference: Toller Rucksack, aber bei den Riemen übertrieben
Translation: Toller Rucksack, aber übertrieben auf den Riemen



Model	Score
Expert	59
GPT-4	90
Claude-3	96
Gemini-1.5	95

Source: LLMs instead of Human Judges? A Large Scale Empirical Study across 20 NLP Evaluation Tasks, 2024

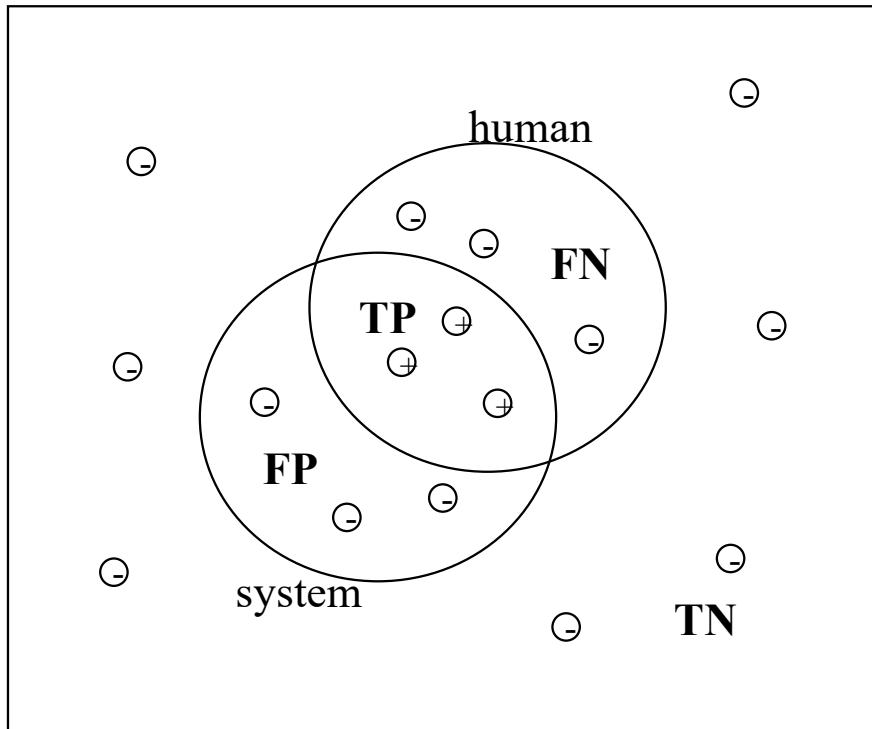
Contingency Matrix (1)

A contingency matrix allows to compare the results (automatically) obtained by a text mining system with the correct judgments (manually) provided by a human on the same dataset.

		Human Judgments (model)	
		Relevant	Not relevant
TM system	Relevant	<i>True-Positive</i>	<i>False-Positive</i>
	Not relevant	<i>False-Negative</i>	<i>True-Negative</i>

Contingency Matrix: Set Model View

Evaluation data



⊕ Relevant document

⊖ Not relevant document

TP: True Positive

TN: True-Negative

FP: False-Positive

FN: False-Negative

Evaluation Metrics

Accuracy: percentage of cases correctly classified by the system.

$$\frac{TP + TN}{TP + TN + FP + FN}$$

Error Rate: Inverse of accuracy. Percentage of cases wrongly classified by the system.

$$\frac{FP + FN}{TP + TN + FP + FN}$$

Evaluation Metrics

- **Accuracy** is not a good measure for all tasks in computational linguistics
- When the number of True Negative (TN) is much bigger than the number of True Positive, False Positive and False Negative accuracy tends to be 1
- TNs make the proportion not relevant.
- We need a measure that does not consider TNs.

Evaluation Metrics

Precision: percentage of yes cases correctly classified by the system with respect to all cases classified by the system.

$$\frac{TP}{TP + FP}$$

Recall: percentage of yes cases correctly classifies by the system with respect to all cases classifies by the human.

$$\frac{TP}{TP + FN}$$

Evaluation Metrics

- **F-Measure:** combine in a single measure Precision (P) and Recall (R), giving a global estimation of the performance of a classifier.

$$F = \frac{1}{\alpha \frac{1}{P} + (1 - \alpha) \frac{1}{R}}$$

- The α parameter is used to tune the contribution of precision and recall. With $\alpha = 0,5$, the formula can be simplified with $2PR/(R + P)$. If α is above 0,5 then precision is rewarded.

Evaluation Metrics

		Human Judge	
		yes	no
Model	yes	$TP = 10$	$FP = 30$
	no	$FN = 5$	$TN = 55$

$$\text{Accuracy} = (10 + 55) / 100 = 65/100$$

$$\text{Error rate} = (5 + 30) / 100 = 35/100$$

$$\text{Precision} = 10 / (10 + 30) = 0,25$$

$$\text{Recall} = 10 / (10 + 5) = 0,66$$

$$\text{F-Measure} = (2 * 0,25 * 0,66) / (0,25 + 0,66) = 0,36$$

LLM-Leaderboard vs LLM-Arena

Ever-evolving LLM Capabilities

LLM performance evolve at high speed

- **Risk of performance saturation:** benchmarks may become obsolete in a short time
- **Linguistic tasks** (e.g., parsing) may appear as less interesting compared to other abilities (e.g., solve math problems with reasoning)
- **Rise the threshold:** develop more difficult tasks/questions (e.g., Humanity's Last Exam, 2025)
- **No Evalita-LLM task seem to be saturated by current mid-size models** (max accuracy = 0.8)

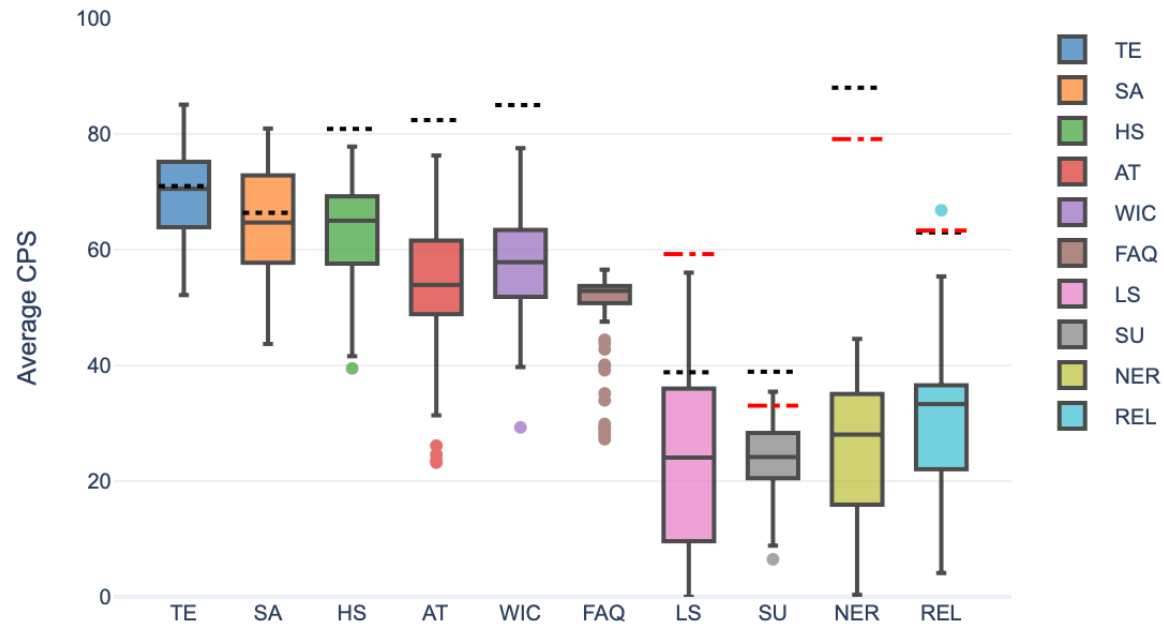
	Meta Llama 3 8B	Gemma 7B - It Measured	Mistral 7B Instruct Measured
MMLU 5-shot	68.4	53.3	58.4
GPQA 0-shot	34.2	21.4	26.3
HumanEval 0-shot	62.2	30.5	36.6
GSM-8K 8-shot, CoT	79.6	30.6	39.9
MATH 4-shot, CoT	30.0	12.2	11.0

How to Evaluate: LLM Leaderbord

Rank ▲	Size ▲	FS ▲	Model ▲	Avg. Comb. Perf. ⬆ ▲	TE ▲	SA ▲	HS ▲	AT ▲	WIC ▲	FAQ ▲	LS ▲
1	●●●	5	Mistral-Large-Instruct-2411 ●●●🏆	62.28	80.96	80.94	77.03	76.28	75.55	54.42	38
2	●●●	5	Llama-3.1-Tulu-3-70B	60.16	82.42	80.41	75.51	72.44	65.44	54.52	40
3	●●●	5	Llama-3.3-70B-Instruct	57.93	80.79	79.55	77.82	70.72	67.15	54.11	46
4	●●	5	Mistral-Small-24B-Instruct-2501 ●●🏆	57.67	82	75.95	72.9	69.77	71.65	53.79	41
5	●●	5	gemma-3-27b-it	57.42	81.13	80.3	75.87	73.91	66.49	53.13	36
6	●●●	5	Qwen2.5-72B-Instruct	57.36	85.07	77.68	76.38	69.3	72.06	54	38
7	●●●	5	calme-3.2-instruct-78b	56.82	84.31	80.15	75.82	64.68	73.44	53.6	38
8	●●●	🔴	Qwen3-Next-80B-A3B-Instruct ●●●🏆	55.35	73.89	67.09	69.84	62.53	71.41	53.41	54
9	●●	5	Qwen2.5-VL-32B-Instruct	55.3	83.7	68.13	75.47	61.53	77.56	52.57	32
10	●●	5	Qwen2.5-14B-Instruct-1M	54.72	85.02	73.17	72.8	59.75	63.59	52.61	35

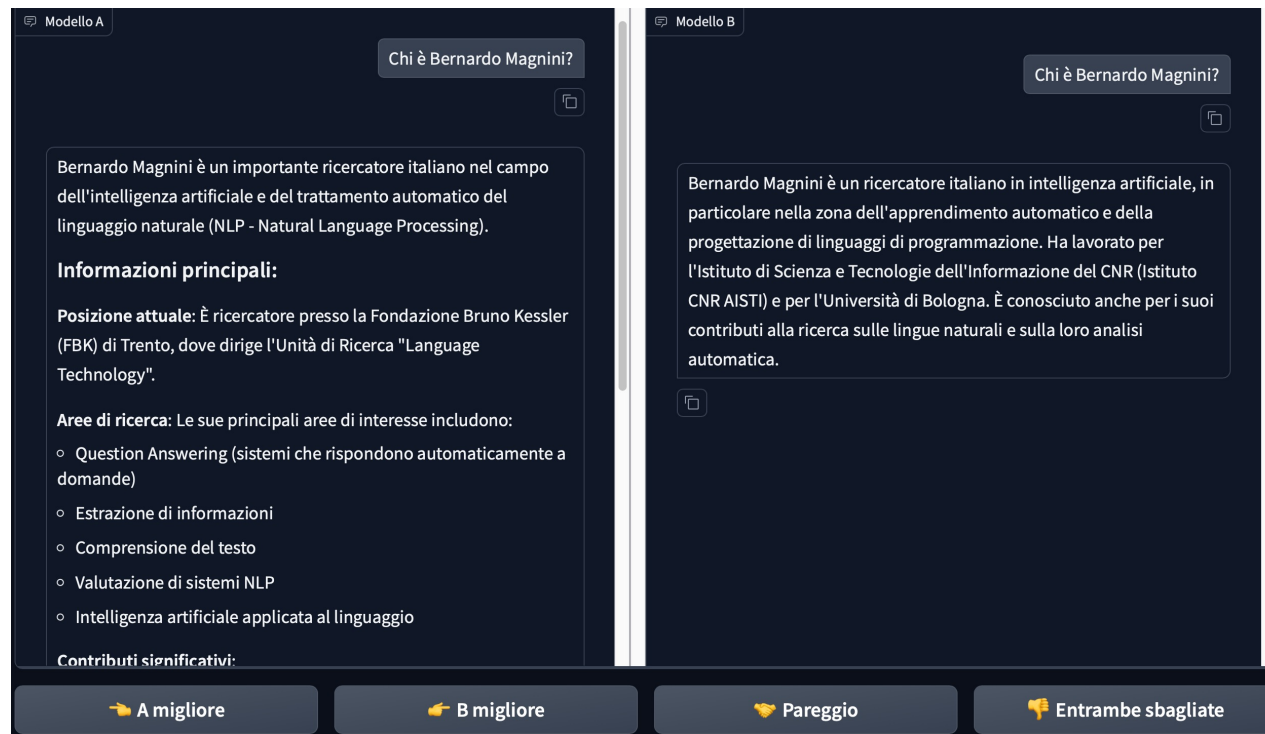
- Ranking is obtained by averaging models on a set of predefined tasks
- Several automatic metrics (including LLM-as-Judge)
- For Italian: Evalita-LLM - https://huggingface.co/spaces/evalitahf/evalita_llm_leaderboard

How to Evaluate: LLM Leaderbord



- Statistics are automatically derived from the ranking (e.g., performance per task)
- For Italian: Evalita-LLM - https://huggingface.co/spaces/evalitahf/evalita_llm_leaderboard

How to Evaluate: LLM-Arena

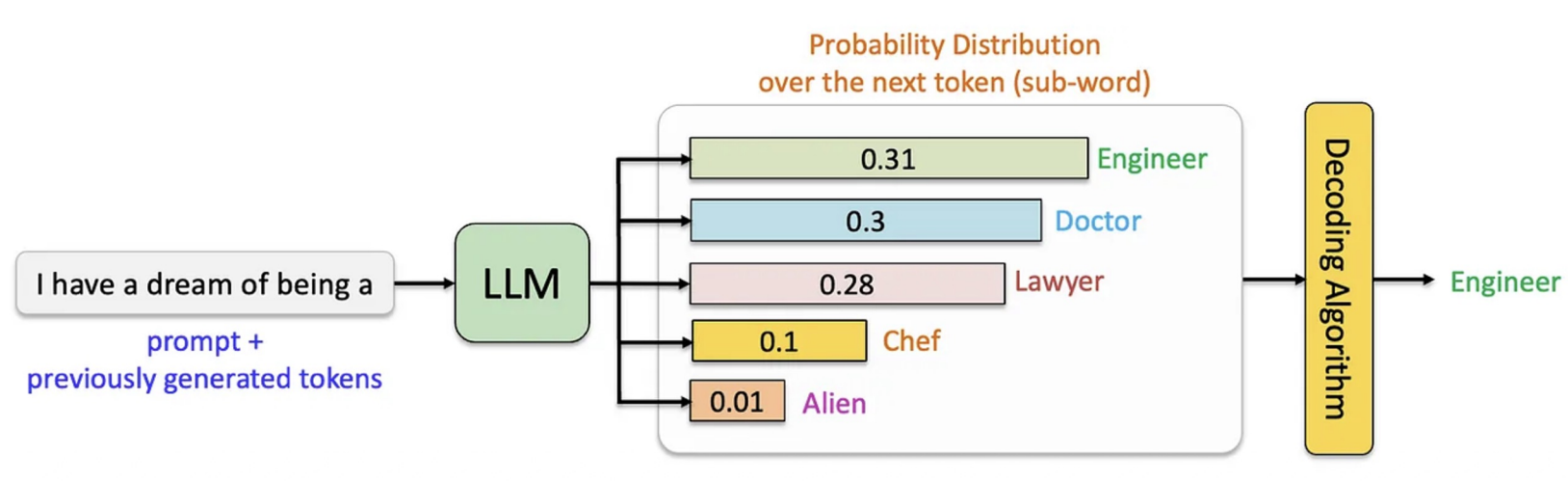


- Ranking is obtained by comparing two models on dynamic tasks (user generated)
- Based on human alignment
- For Italian: Chatbot Arena <https://indigo.ai/it/chatbot-arena-italia/>

Single Prompt vs Multiple Prompts

Let's Zoom-in...

Probability distribution is sensitive to prompts

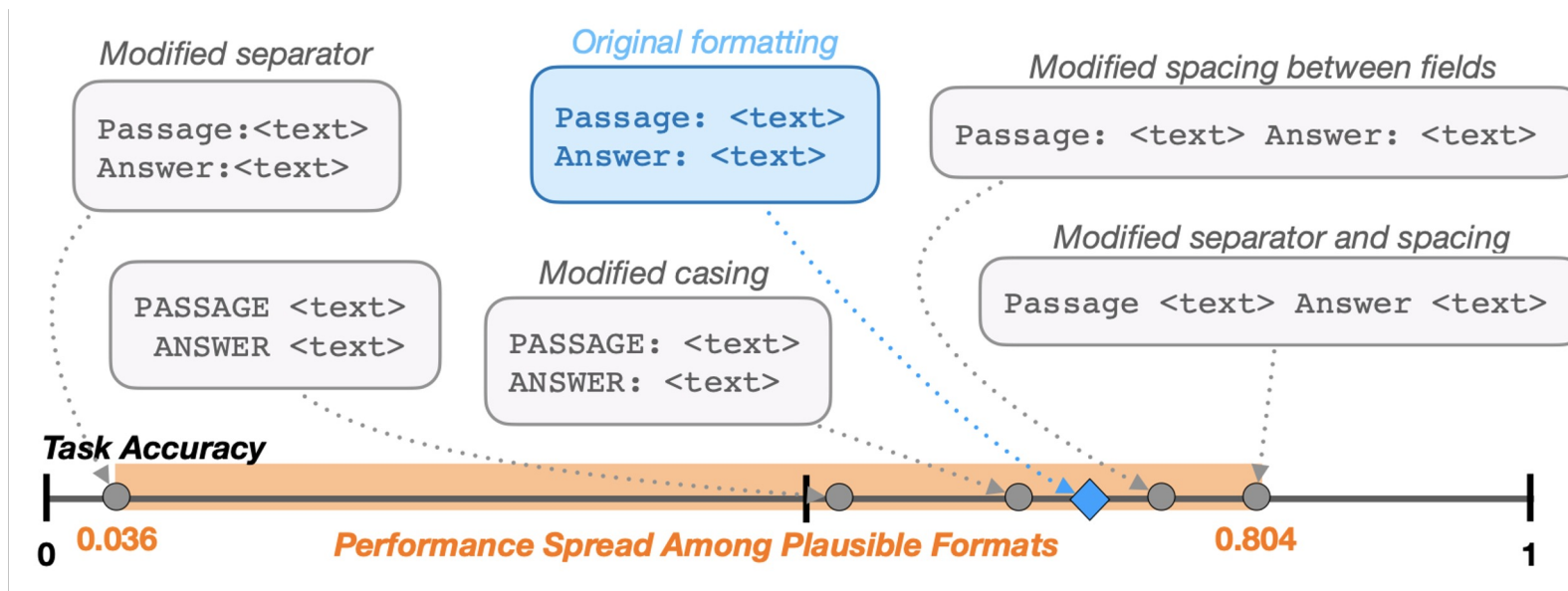


Source: <https://medium.com/@lmpo/mastering-llms-a-guide-to-decoding-algorithms-c90a48fd167b>

Prompt Sensitivity: Formatting

LLMs (particularly small-medium) are very sensitive to prompt variations

- Same model, different prompts
- Different models, same prompt

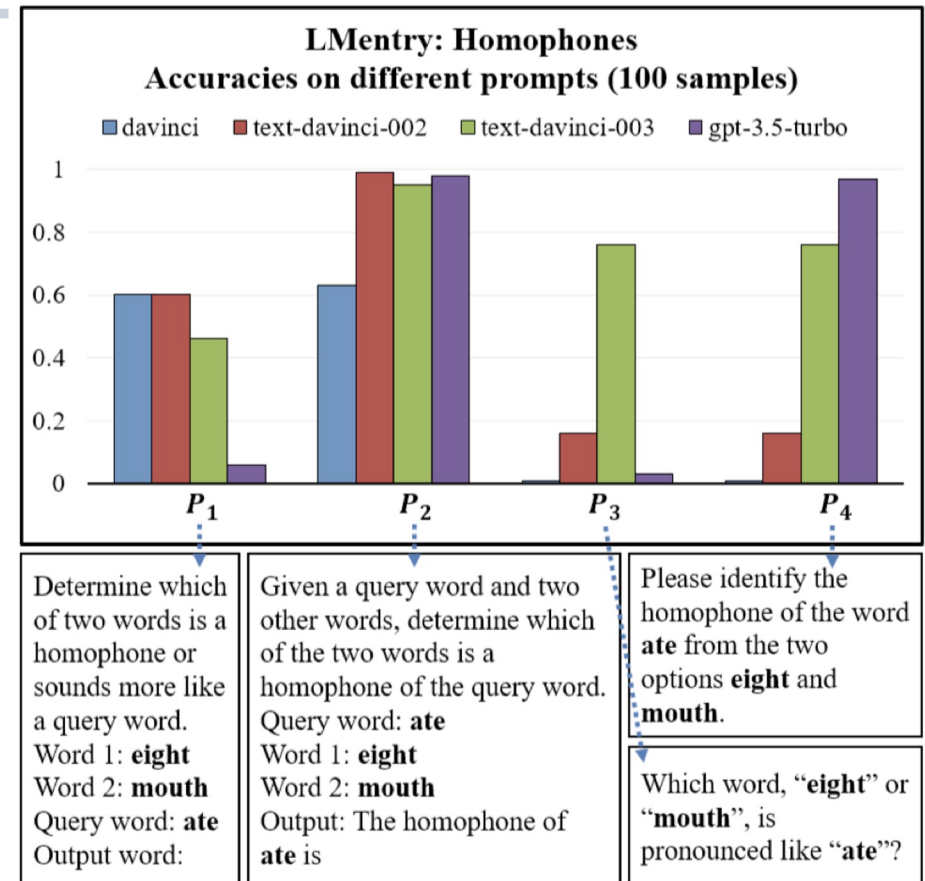


Sclar et al.: Quantifying Language Models' Sensitivity to Spurious Features in Prompt Design or: How I learned to start worrying about prompt formatting, ICLR 2024.

Prompt Sensitivity: Paraphrasing

LLMs (particularly small-medium) are very sensitive to prompt variations

- Paraphrase variations of a prompt results in significant differences in accuracy
- Different models react differently to prompt variations (e.g., P1 is good for LLM 1, but poor for LLM 4)
- Difficult to predict why: base vs instructed models
- Prompt sensitivity can be mitigated adopting a **multiple prompt approach**



Best Practices for Evaluation Prompting

Few general practical rules

- **Language.** Do not mix languages (e.g., Italian only).
- **Role of the model.** As different LLMs might be instructed to converse with specific prompts, it is preferable not to use roles.
- **Minimality.** Use a simple and as short as possible wording. Minimally verbose prompts are preferred in order to reduce potential biases for a model.
- **Input text.** Specify the type of input text required by a task. For instance, ask the model to perform a certain task on a tweet, news, sentence, words.
- **Prompt Wizard (e.g.,: Microsoft)**

Source: <https://it.linkedin.com/pulse/il-potere-del-prompting-la-competenza-chiave-per-futuro-facchin>



Prompt Composition

Composing prompts from basic elements

- **Question (M):** *Which is the sentiment of tweet [tweet]?*
- **Statement (M):** *The following tweet [tweet] expresses a sentiment*
- **Role (O):** *You are an expert linguist.*
- **Task description (O):** *This is a sentiment analysis task.*
- **Answers (O):** A: positive B: negative C: neutral D: mixed

E.g. Task description + question + answers

This is a sentiment analysis task. Which is the sentiment of tweet [tweet]?

A: positive B: negative C: neutral D: mixed

Answer:

Prompt Composition (Evalita-LLM)

	Composition	Prompt	Choices
1	Question	Which is the sentiment of tweet '{{text}}'?	["Positive", "Negative", "Neutral", "Mixed"]
2	Question + task description	This is a sentiment analysis task. Which is the sentiment of tweet '{{text}}'?	["Positive", "Negative", "Neutral", "Mixed"]
3	Question + answer	Which is the sentiment of tweet '{{text}}'? A: Positive B: Negative C: Neutral D: Mixed Answer:	["A", "B", "C", "D"]
4	Question + answer + task description	This is a sentiment analysis task. Which is the sentiment of tweet '{{text}}'? A: Positive B: Negative C: Neutral D: Mixed Answer:	["A", "B", "C", "D"]
5	Statement	The following tweet: '{{text}}' expresses a sentiment	["Positive", "Negative", "Neutral", "Mixed"]
6	Statement + task description	This is a sentiment analysis task. The following tweet: '{{text}}' expresses a sentiment	["Positive", "Negative", "Neutral", "Mixed"]

Multiple Prompts: Metrics

Maximum Performance: useful for downstream applications

$$MaxP(M, T, I_T) = \max_{i \in I_T} \varepsilon(M, T, i)$$

M = model (LLM)
T = task
I _T = set of prompts

Average Performance: useful to assess model's robustness

$$AvgP(M, T, I_T) = \frac{1}{|I_T|} \cdot \sum_{i \in I_T} \varepsilon(M, T, i)$$

Combined performance score: useful for a model used for different tasks

$$Sat(M, T, I_T) = 1 - (MaxP - AvgP)$$

$$CPS(M, T, I_T) = Sat \cdot MaxP$$

Mizrahi et al.: State of What Art? A Call for Multi-Prompt LLM Evaluation, 2024.

Multiple Prompts (Evalita-LLM)

	LLM 1	LLM 2	LLM 3	LLM 4	LLM 5	LLM 6	
P1	33,06	43,9	36,93	52,99	41,14	41,19	
P2	37,19	49,35	36,71	38,69	59,4	49,82	
P3	53,33	64,18	43,24	26,27	65,64	69,72	
P4	54,35	68,31	46,74	28,31	70,73	72,17	
P5	53,28	71,42	53,26	62,34	65,68	67	
P6	59,26	63,49	59,06	63,08	68,5	66,2	
MaxP	59,26	71,42	59,06	63,08	70,73	72,17	
AvgP	48,41	60,11	45,99	45,28	61,85	61,02	
CSP	0,528	0,633	0,513	0,519	0,644	0,641	
R	4	3	6	5	1	2	

MaxP	AvgP	CSP	R
52,99	41,54	0,469	6
59,4	45,19	0,510	5
69,72	53,73	0,586	4
72,17	56,77	0,611	3
71,42	62,16	0,648	2
68,5	63,27	0,649	1

Best result

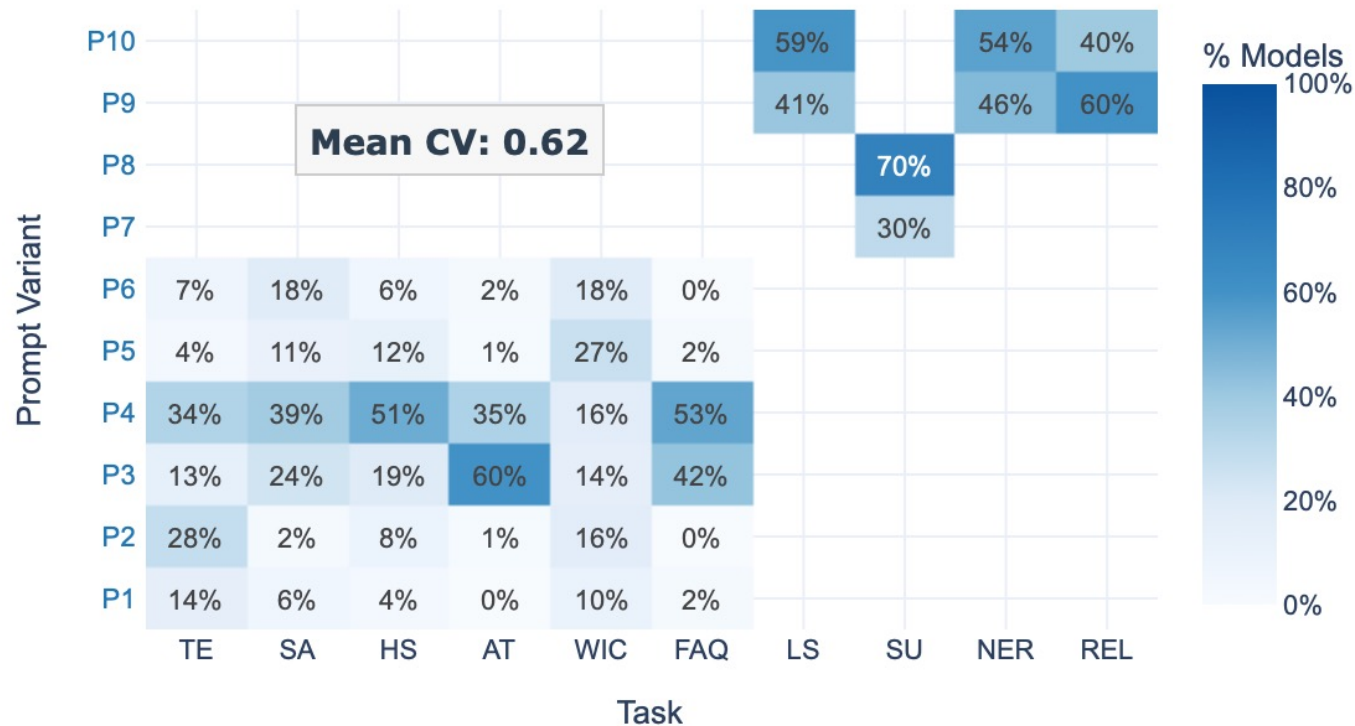
Best average

Best comb.

Evalita-LLM Benchmark

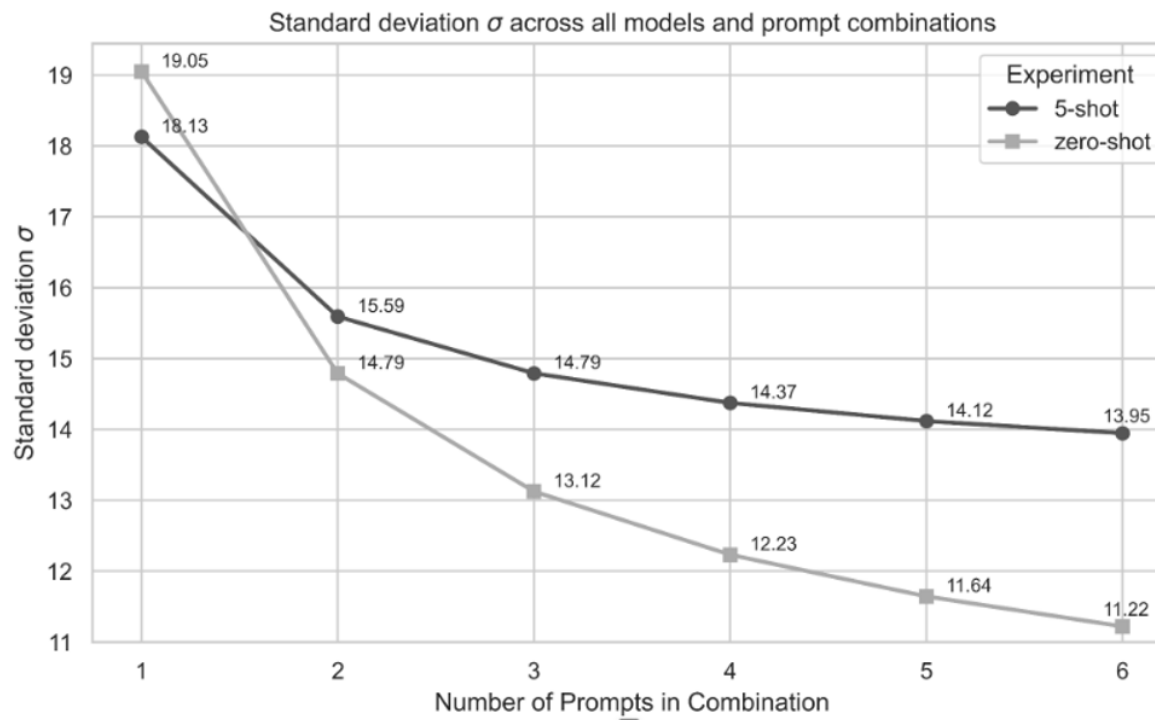
Task = sentiment analysis **Metric** = F1 macro **Configuration** = 0-shot

Multiple Prompts (Evalita-LLM)



Evalita-LLM Benchmark: all 10 tasks. 0-shot and 5-shot, Multiple choice and open-ended
No single prompt achieve better results on all tasks.

Multiple Prompts (Evalita-LLM)



Evalita-LLM Benchmark: all 10 tasks. 0-shot results in a 42% prompt sensitivity from 1 to 6 prompt used.

Multiple-choice vs Open-ended

LLMs: How to Evaluate

Prevalent Approach: **Multiple Choice Questions**

Evaluation is straightforward

MMLU (Massive Multitask Language Understanding - 2021)

- Multiple choice questions: one correct answer, three distractors
- 57 topics: economics, medicine, social science, math, etc.
- 15908 questions (1540 for dev, 14079 for test)
- Metric: accuracy of correct answers

Microeconomics

One of the reasons that the government discourages and regulates monopolies is that

- (A) producer surplus is lost and consumer surplus is gained.
- (B) monopoly prices ensure productive efficiency but cost society allocative efficiency.
- (C) monopoly firms do not engage in significant research and development.
- (D) consumer surplus is lost with higher prices and lower levels of output.



Question Answering Tasks

Three implementations are possible with LLMs:

Multiple choice (MC) + LLM probability

- **Prompt:** *Are there broccoli in the picture?*

Answers: "Yes" "No"

Compare log-likelihoods $\log P(y_i | x)$ across a fixed set of answer strings y_i and select $\text{argmax}_i(\log P(\text{answer}_i | \text{prompt}))$

Multiple choice (MC) + LLM generation

- **Prompt:** *Are there broccoli in the picture? Answer "Yes" or "No".*
 - *Yes, they are in the bowl*

Parse LLM output for **answer_i** and return it

Open-ended (OE) + LLM generation

- **Prompt:** *Are there broccoli in the picture? Answer:*

- *There are two broccoli in the picture*

If a reference answer is available use automatic metrics (rouge, bert-score, etc.), otherwise human judgement



Source: Pope visual QA benchmark - 2023

Confronting MC-prob and MC-gen

Same classification task: sentiment analysis

Prompt: This is a sentiment analysis task. Which is the sentiment of tweet '{{text}}'? A: Positive B: Negative C: Neutral D: Mixed
Answer:

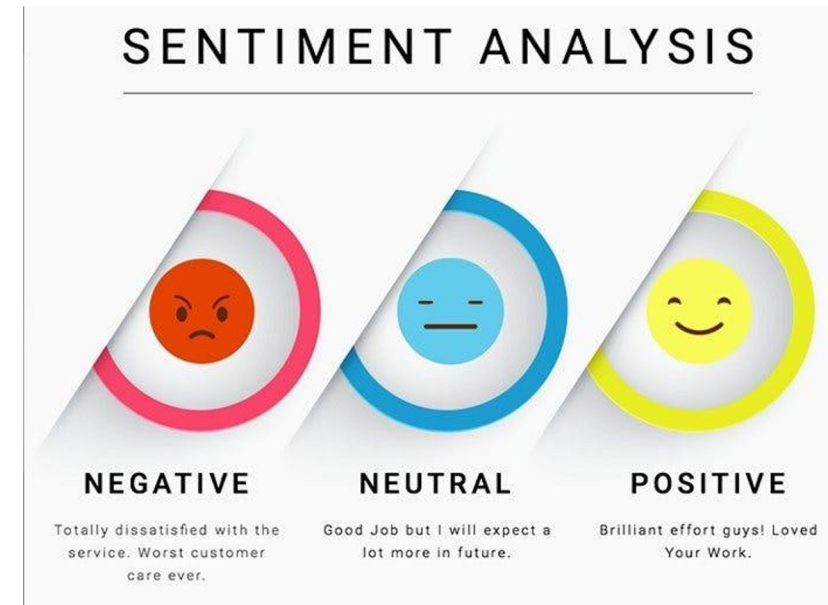
Multiple choice + probability

- Gemma 9B: 70,7 F1
- Velvet 14B: 60,49 F1

Multiple choice + answer generation

- Gemma 9B: 70,0 F1
- Velvet 14B: 64,20 F1

Similar results for Gemma, significant improvement for Velvet



Source: <https://blog.gopenai.com/sentiment-analysis-on-healthcare-reviews-2ec229d04e69>

Open-ended Questions

Test the generative ability of LLMs

E.g., summarization, open question answering

Certain tasks only have a **generative formulation**:

- Prompt: *You have to solve a task of automatic summarization. Summarize the following text: '{{source}}'\nSummary:*
- Typically, the LLM output is compared against **human references**
- Several automatic metrics (e.g., rouge, bert-score) are not adequate for generative tasks
- **LLM-as-a-Judge as a new evaluation paradigm**



Source: ChatGPT 2025.

Confronting MC and OE

Example from the MMLU-ita dataset

Multiple choice + generation

Un aumento della
frequenza di minzione
negli adulti anziani è molto
probabilmente causato da:

- A) cancro alla prostata.
- B) **Vescica piu' piccola**
- C) Problemi renali
- D) Calcoli biliari

Open Ended

Un aumento della
frequenza di minzione
negli adulti anziani è molto
probabilmente causato da:
Rispondi

Confronting MC and OE Formulations

MC + generation: Gemma 3 4B: wrong answer

La risposta corretta è ****C: problemi renali.****

Ecco perché:

- * ****Problemi renali:**** **Molte condizioni renali possono causare un aumento della frequenza di minzione.** I reni regolano l'equilibrio dei liquidi nel corpo, e se non funzionano correttamente, questo può portare a minzione più frequente.
- * ****Cancro alla prostata (A):**** Il cancro alla prostata negli uomini può causare problemi di minzione, ma più spesso si manifesta con difficoltà a urinare, flusso debole o gocciolamento, piuttosto che con un aumento della frequenza.
- * ****Vescica più piccola (B):**** Anche se la vescica può cambiare con l'età, **una riduzione significativa delle dimensioni è meno probabile causa primaria di un aumento della frequenza di minzione.**
- * ****Calcoli biliari (D):**** I calcoli biliari riguardano il fegato e la cistifellea e non hanno alcun impatto sulla frequenza di minzione.

****Importante:**** Un aumento della frequenza di minzione dovrebbe sempre essere valutato da un medico per escludere cause sottostanti.

Confronting MC and OE Formulations

OE + generation: Gemma 3 4B: better answer

Un aumento della frequenza di minzione negli adulti anziani è molto probabilmente causato da una combinazione di fattori, tra cui:

* **Debolezza dei muscoli del pavimento pelvico:** Questi muscoli supportano la vescica e l'uretra. Con l'età, possono indebolirsi, portando a una **vescica meno efficiente**.

* **Riduzione della funzione renale:** I reni potrebbero non essere in grado di concentrare l'urina in modo efficiente, portando a una maggiore produzione di urina.

* **Condizioni mediche:** Diabete, infezioni delle vie urinarie, ipertrofia prostatica benigna (negli uomini), incontinenza urinaria e altre condizioni possono contribuire.

* **Farmaci:** Alcuni farmaci, come i diuretici, possono aumentare la frequenza di minzione.

* ****Ridotta capacità della vescica:**** La vescica può perdere elasticità con l'età.

* **Consumo di liquidi:** Anche se non è una causa diretta, l'assunzione di liquidi, soprattutto bevande diuretiche come caffè e tè, può esacerbare il problema.

* **È importante consultare un medico per una diagnosi accurata e un trattamento appropriato, poiché un aumento della frequenza di minzione può essere un sintomo di una condizione medica sottostante.**

Open-ended Tasks

The answer does not belong to a set of predefined labels

E.g., summarization, open question answering, translation, etc.

Only the **generative formulation** is possible:

- Prompt: *You have to solve a task of automatic summarization. Summarize the following text:*
'{{source}}'\nSummary:
- Typically, evaluate the LLM output by comparison with **human references**.
- Several metrics: rouge, bert-score, blaser, etc.
- **LLM-as-a-Judge** also possible

Please read the passage

This has been applied mainly for text. Abstractive methods build an internal semantic representation of the original content, and then use this representation to create a summary that is closer to what a human might express. Abstraction may transform the extracted content by paraphrasing sections of the source document, to condense a text more strongly than extraction. Such transformation, however, is computationally much more challenging than extraction.

Provide one sentence summary:

There are two approaches to automatic summarization: extraction and abstraction. In extraction summarization content is extracted from the original data, whereas Abstraction may transform the extracted content by



Source: https://labelstud.io/templates/text_summarization

Combining MC+gen with OE Tasks

Classification: caption/foil

Open-ended: question answering

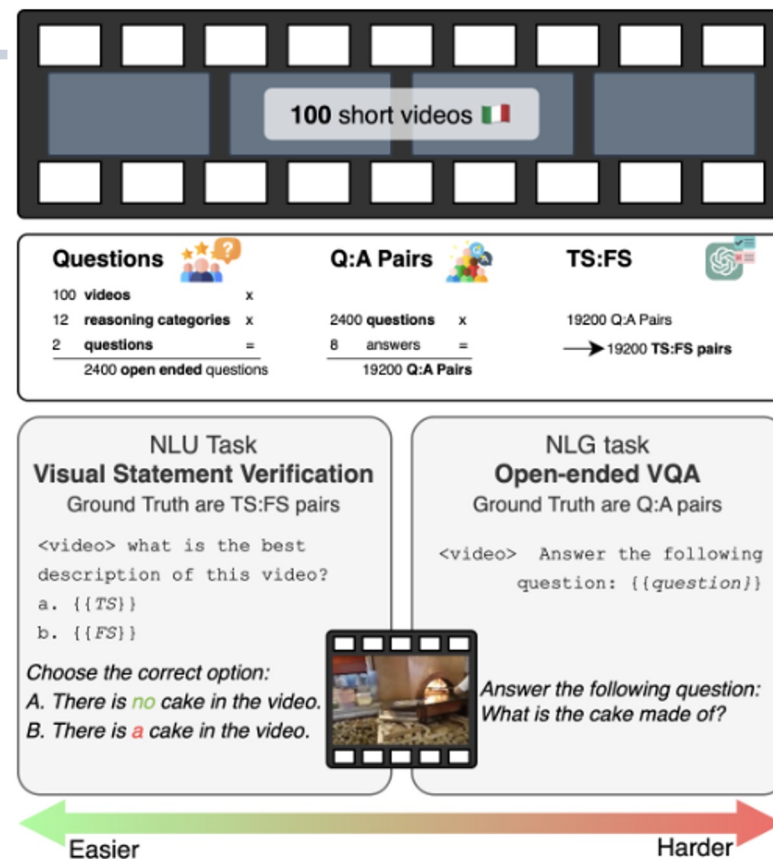
MAIA video question answering (32 frames per video, each 30')

Same question for the two modalities.

Qwen-2.5-VLI:

- Caption-foil: Accuracy = 0.84
- Open-ended: Accuracy = 0.61

Significant performance decrease for open-ended questions



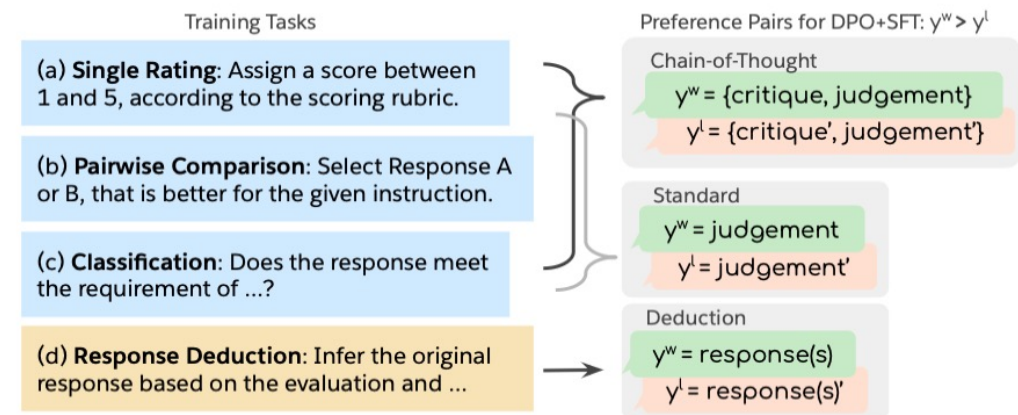
Source: MAIA visual QA benchmark - 2025

LLM-as-Judge

LLM_as_Judge

LLM_as_Judge: three approaches

- **Prompting high performing models**, training specialized judge models
- **Supervised Fine Tuning is suboptimal**, as it only trains LLMs to mimic correct examples without exposing the model to incorrect examples
- **Direct Preference Optimization (DPO)**
preference data: Chain-of-thought Critique, Standard Judgement and Response Deduction



LLM_as_Judge

Input

$$x = (p, i, r) \in X$$

where $p \in P$ is an **evaluation protocol** task description (single rating, pairwise, or classification) and an evaluation rubric, which specifies the rules and criteria for evaluation (e.g., helpfulness, safety, etc.)

$i \in I$ is a **task input**, user input (prompt to the model)

$r \in R$ is a set of **model responses**: single response $\{r\}$ or a pair of responses $\{r1, r2\}$

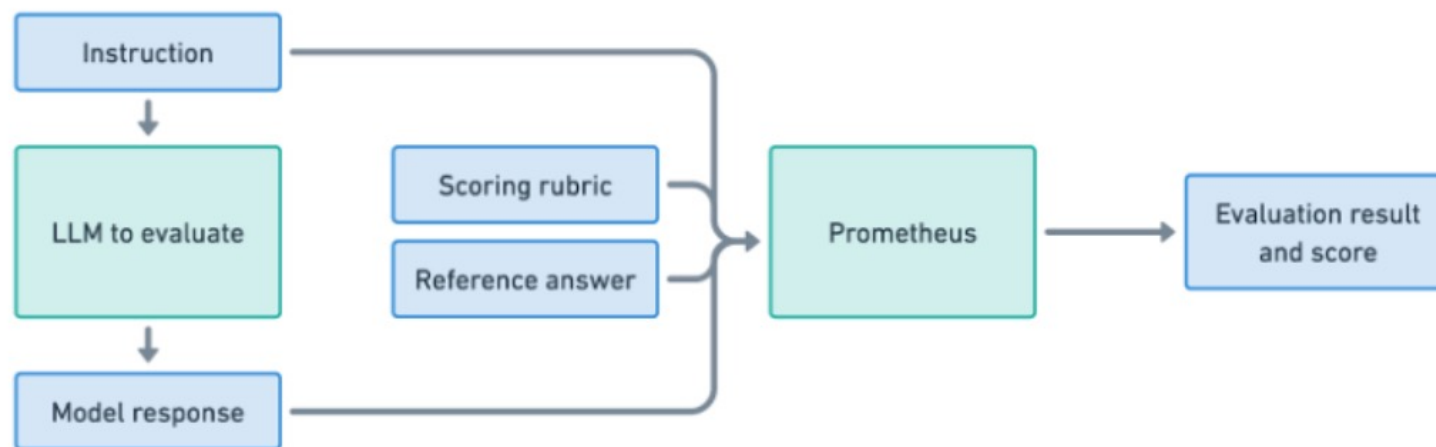
Output

Free-text evaluation $y \in Y, \{c, j\}$, where c is a natural language critique/explanation and j is the model's judgement

LLM_as_Judge: Prometheus

LLM_as_Judge paradigm

- Parwise evaluation (< 100K training samples)
- Single-instance rating, pairwise comparison, binary classification, and response deduction
- DPO preference data: Chain-of-thought Critique, Standard Judgement and Response Deduction

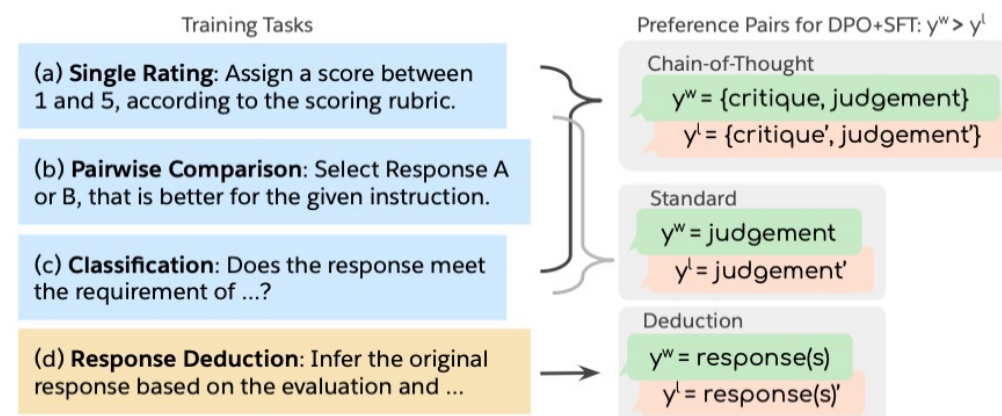


LLM_as_Judge: Training

LLM_as_Judge paradigm

Training a judge requires training datasets of (x, y) pairs, where y is an evaluation that consists of a critique c^* and final judgment j^* produced by either humans or frontier LLMs.

- Parwise evaluation ($< 100K$ training samples)
- Single-instance rating, pairwise comparison, binary classification, and response deduction



More issues...

Benchmarking LLMs: an Active Research Area

- Data contamination
- Ever-evolving capabilities
- Static vs dynamic benchmarks
- Sustainability of LLM evaluation

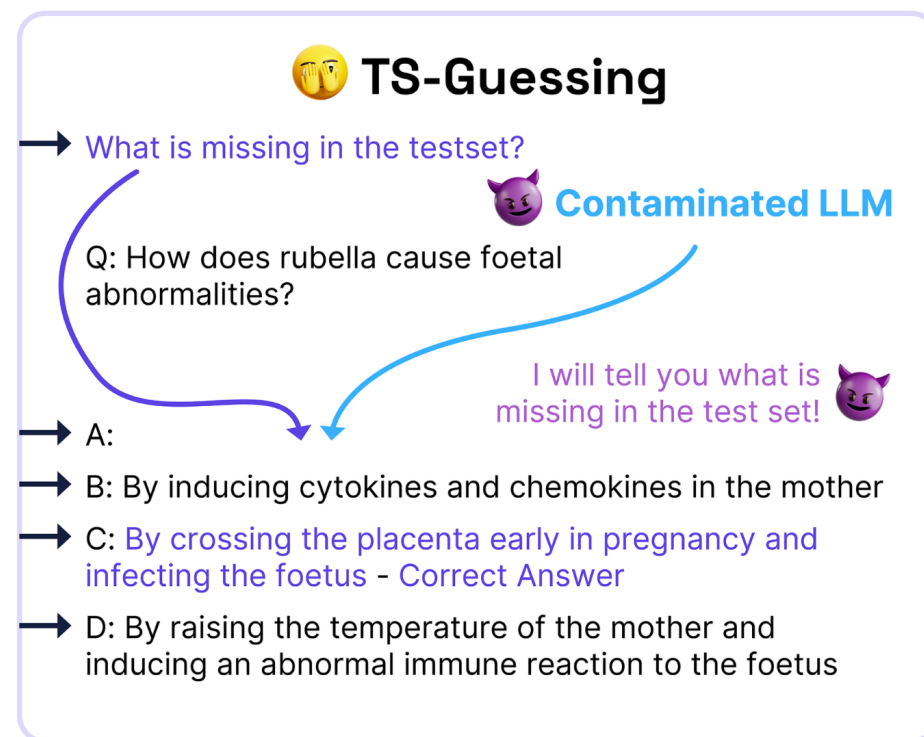


Source: ChatGPT 2025.

Data Contamination

Potential contamination of LLMs due to massive pre-training

- May cause unfair evaluation
- Develop method to detect potential data contamination
- Probing the LLM to generate test data
- GPT-4 demonstrated an exact match rate of 57% in TS-guessing
- Results on Evalita-LLM seem to **exclude contamination of Italian datasets**



Source: Denget al: Investigating Data Contamination in Modern Benchmarks for Large Language Models, Naac1, 2024.

Dynamic Settings

Currently evaluation addresses **static tasks**.

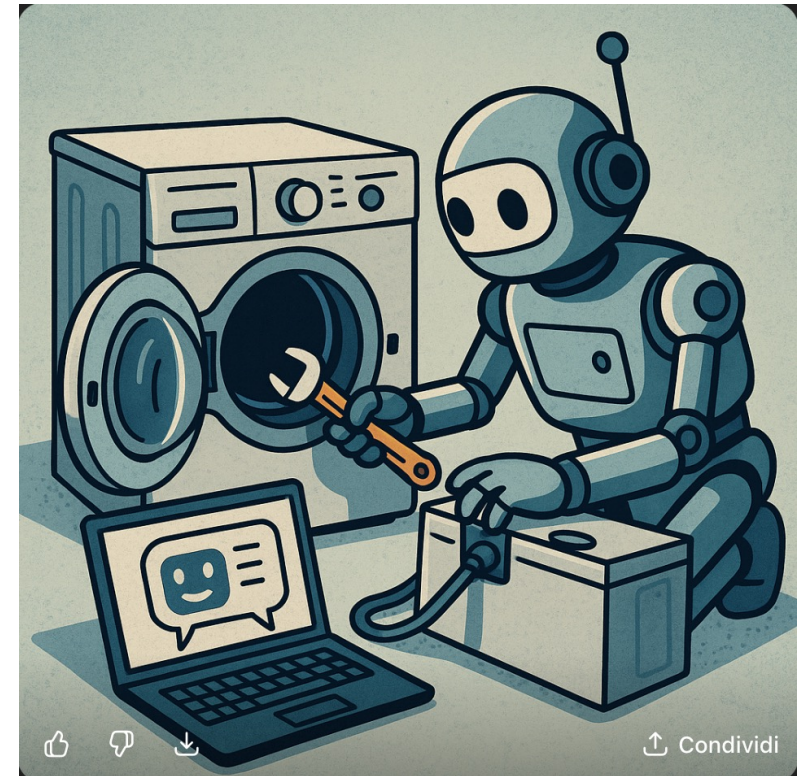
Dynamic tasks (e.g., dialogue, planning) can not be foreseen

Robot: *Come posso staccare la corrente?*

LLM: Verifica l'alimentatore di corrente.

Robot: *Ho capito, ma dove trovo l'alimentatore?*

- **Environment is changing**, new questions need to be formulated for new situations
- **Simulated environments** may help (e.g., block world, house rooms)
- **LLM as a judge** as a possible solution



Source: ChatGPT 5, September 2025.

Conclusion

Rethinking NLP Evaluation

Fair evaluation of LLMs is still challenging

- LLM-leaderboard vs LLM-Arena
- Single prompt vs multiple prompts
- Multiple-choice vs open-ended tasks
- LLM-as-Judge
- More issues: data contamination, dynamic tasks, sustainability

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